

OPDE

Linear Partial Differential Equation of order one:-
(Lagrange's Method).

Lagrange's Equation:- An equation of the form $Pp + Qq = R$ or $P \frac{\partial z}{\partial x} + Q \frac{\partial z}{\partial y} = R$ where P, Q, R are function of x, y, z is called Lagrange's equation.

Lagrange's Auxilliary Equation:

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

* General soln is: $\phi(u, v) = 0$

where ϕ is an arbitrary function

& $u = C_1, v = C_2$

C_1, C_2 are integral Constant.

$$1. y^2 p - xy q = xz$$

$$2. p + 3q = 5z + \tan(y - 3x)$$

$$3. p - 2q = 3x^2 \sin(y + 2x)$$

$$4. y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y} = xyz^2(x^2 - y^2)$$

$$5. xy \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = zx^2 - 2x^2$$

$$6. (mz - ny) \frac{\partial z}{\partial x} + (nx - lz) \frac{\partial z}{\partial y}$$

$$= ly - mx.$$

$$7. x(y^2 - z^2)p - y(x^2 + z^2)q = z(x^2 + y^2).$$

$$8. z(x + y) \frac{\partial z}{\partial x} + z(x - y) \frac{\partial z}{\partial y} = x^2 + y^2.$$

Soln:-1 Given,

$$y^2 p - xy q = xz$$
$$y^2 \frac{\partial z}{\partial x} - xy \frac{\partial z}{\partial y} = xz \quad \text{--- (i)}$$

Here,

$$P = y^2$$

$$Q = -xy$$

$$R = xz$$

Lagrange's auxiliary equation are,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{xz} \quad \text{--- (ii)}$$

Taking the first two part from (ii),

$$\therefore \frac{dx}{y^2} = \frac{dy}{-xy}$$

$$\Rightarrow \frac{dx}{x} = \frac{dy}{-y}$$

$$\Rightarrow x dx + y dy = 0$$

$$\therefore \int x dx + \int y dy = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} = \frac{C_1}{2}$$

$$\therefore x^2 + y^2 = C_1$$

Again taking the last

$$\frac{dy}{xy} = \frac{dz}{xz}$$

$$\Rightarrow \frac{dz}{z} = \frac{dz}{-z}$$

$$\therefore \int \frac{1}{y} dy + \int \frac{1}{z} dz = 0$$

$$\Rightarrow \ln y + \ln z = \ln C_2$$

$$\therefore yz = C_2$$

$$\therefore u = C_1 = x^2 + y^2$$

$$v = C_2 = yz$$

∴ General Soln is:

$$\phi(u, v) = 0$$

$$\therefore \phi(x^2 + y^2, yz) = 0$$

Ans

Soln:-3 Given,

$$P - 2Q = 3x^2 \sin(y + 2x) \quad \text{--- (i)}$$

$$\text{Hence, } P = 1$$

$$Q = -2$$

$$R = 3x^2 \sin(y + 2x)$$

Lagrange's auxiliary equation are,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{1} = \frac{dy}{-2} = \frac{dz}{3x^2 \sin(y + 2x)} \quad \text{--- (ii)}$$

Taking the first two part from (i)

$$\therefore \frac{dx}{1} = \frac{dy}{-2}$$

$$\Rightarrow dy = -2 dx$$

$$\therefore \int dy + 2 \int dx = 0$$

$$\therefore y + 2x = C_1 \quad \text{--- (iii)}$$

Again taking the first & last parts from (ii).

$$\therefore \frac{dx}{1} = \frac{dz}{3x^2 \sin(y+2x)}$$

$$\Rightarrow 3x^2 dx = \frac{dz}{\sin C_1} \quad [\text{using (iii)}]$$

$$\therefore 3 \sin C_1 \int x^2 dx + \int dz = 0$$

$$\Rightarrow 3 \sin C_1 \cdot \frac{x^3}{3} + z = C_2$$

$$\therefore x^3 \sin(y+2x) - z = C_2$$

$$\therefore u = y+2x$$

$$v = x^3 \sin(y+2x) - z$$

$$\therefore \text{General soln. is } \phi(u, v) = 0$$

$$\therefore \phi \{ y+2x, x^3 \sin(y+2x) - z \} = 0$$

Ans

Soln :- 6 Given,

$$(mz - ny)p + (nx - lz)q = ly - mx \quad \text{--- (i)}$$

The Lagrange's auxiliary equation are,

$$\frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx} \quad \text{--- (ii)}$$

Choosing x, y, z as multipliers,
each fraction of (ii),

$$\Rightarrow \frac{x dx + y dy + z dz}{x(mx - ny) + y(ny - lz) + z(ly - mx)} \\ = \frac{x dx + y dy + z dz}{0}$$

$$\therefore x dx + y dy + z dz = 0$$

$$\therefore \int x dx + \int y dy + \int z dz = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{C_1}{2}$$

$$\therefore x^2 + y^2 + z^2 = C_1$$

Again, choosing multipliers l, m, n
each...

$$\Rightarrow \frac{l dx + m dy + n dz}{l(mx - ny) + m(ny - lz) + n(ly - mx)} \\ = \frac{l dx + m dy + n dz}{0}$$

$$\Rightarrow lx + my + nz = 0$$

$$\therefore \int dx + m \int dy + n \int dz = 0$$

$$\therefore lx + my + nz = C_2$$

$\therefore$ General soln is:

$$\phi(u, v) = 0$$

$$\therefore \phi(x^2 + y^2 + z^2, lx + my + nz) = 0 \quad \underline{\underline{Ans}}$$

note:

[By a well known principle of algebra each fraction in (ii) will be equal to $\frac{P_1 dx + Q_1 dy + R_1 dz}{P_1 P + Q_1 Q + R_1 R}$ — (ii)]

If $P_1 P + Q_1 Q + R_1 R = 0$, then we know that the numerator (ii) will be zero, Hence P_1, Q_1, R_1 are called multipliers].



Soln:-7 Given,

$$x(y^2 - z^2)P - y(z^2 + x^2)Q = z(x^2 + y^2)R$$
$$\frac{dx}{x(y^2 - z^2)} = \frac{dy}{-y(z^2 + x^2)} = \frac{dz}{z(x^2 + y^2)}$$

multipliers: x, y, z &

$$\frac{1}{x}, -\frac{1}{y}, \frac{1}{z}$$

(বাকিটা নিজে কর).

Soln:-8 Given,

$$z(x+y)P + z(x-y)Q = x^2 + y^2R$$
$$\frac{dx}{z(x+y)} = \frac{dy}{z(x-y)} = \frac{dz}{x^2 + y^2}$$

multipliers: $x, -y, -z$ &

$$y, x, -z$$

(নিজে নিজে নিয়ে কর বাকিটা).

